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REGULAR PAPER



A demand forecasting system of product categories defined by their time series using a hybrid approach of ensemble learning with feature engineering

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Abstract

Retail companies face major problems in the estimation of their product's future demand due to the high diversity of sales behavior that each good presents. Different forecasting models are implemented to meet the demand requirements for efficient inventory management. However, in most of the proposed works, a single model approach is applied to forecast all products, ignoring that some methods are better adapted for certain features of the demand time series of each product. The proposed forecasting system addresses this problem, by implementing a two-phase methodology that initially clusters the products with the application of an unsupervised learning approach using the extracted demand features of each good, and then, implements a second phase where, after a feature engineering process, a set of different forecasting methods are evaluated to identify those with best performs for each cluster. Finally, ensemble machine learning models are implemented using the top-performing models of each cluster to carry out the demand estimation. The results indicate that the proposed forecasting system improves the demand estimation over the single forecasting approaches when evaluating the R^2, MSE, and MASE quality measures.

Keywords Demand forecasting · Machine learning · Feature engineering · Time series · Ensemble learning methods

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1 Introduction

Demand forecasting plays a central role in a successful supply chain plan and the service levels offered to customers. Most of the proposals to estimate demand are made from a single forecast model, ignoring the different consumption behaviors of the offered products.

1.1 Problem statement

Retail companies are characterized by the high variety of products that they offer. This situation forces these companies to develop complex processes that depend on supply plans based on demand forecasts. Any forecast error can be detrimental to this sector, so it must be as accurate as possible [17]. Forecasts considerably higher than actual demand will impact higher capital flow and inventory management costs, while lower forecasts will lead to lower service levels [17]. On the other hand, the articles/products show very different supply times and consumption behaviors. Thus, the daily forecast of the sale of each product becomes a challenging task, characterized by different attributes defined in its time series (TS) [36]. Then again, the categorization of demand patterns facilitates the definition of forecasting methods for product clusters. However, very limited attention has been carried out in the literature [30]. In general, there is a large number of articles that have used different machine learning (ML) approaches to forecast demand [21] but a few studies have been focused on evaluating the model quality, considering clusters of products according to the demand features evidenced in the TS [30], or using ensemble learning techniques in this context [7, 20, 33, 36]. There are initial works that analyze forecasting methods that are better adapted to some types of data, such as the work [9, 15–17], which analyzed several forecasting methods (and combinations of them), with seven features of TS (seasonality, trend, cycle, randomness, number of observations, interval between demands and coefficient of variation). Also, there are initial works that propose an automated method to obtain weighted forecast combinations using TS features, such as [3, 8, 13, 22, 27, 28, 32, 34, 35, 37] approaches.

1.2 Contribution

The problem with the previous works is that they do not do an exhaustive analysis of the TS that describes the behavior of the products, or they do not carry out an exhaustive process of feature engineering for the specific TS that describes the products. This paper proposes a TS feature engineering process to determine product clusters, ensemble learning methods to predict the behavior of each product category, and an autonomous prediction system that exploits the ensemble learning methods to predict a product in real time. Thus, this paper proposes a demand forecasting system for the retail sector, from a product clustering approach based on a TS feature engineering process, using ensemble learning techniques. Specifically, it evaluates several forecasting models based on ML techniques, from a perspective of clustersof products, according to the behavior of their demand TS, using a feature engineering process.

The main contribution of this work is that a reference framework is proposed for the forecast of product demand, which:

- Perform an autonomic product clustering approach based on the demand behavior, using a feature engineering process for TS;
- Implement forecasting models based on ML techniques for each of the previously clustered products;
- Implement ensemble-learning models for an autonomous demand forecasting system by product category.

2 Related work

In this section are presented several works about feature engineering processes for TS, statistical and ML techniques in TS forecasting, ensemble learning models for demand estimation, and demand forecasting models for TS by product categories.

In TS, Christ, et al. [6] highlight that the feature engineering process plays an important role during the early phases of data science projects since it is a process that allows extracting and exploring different variables to evaluate their importance in predicting the desired objective. Some articles have developed proposals for the selection of features in TS in different contexts. For example, Jiménez, et al. [19] developed a hybrid audio descriptor extraction scheme, based on principles of statistical measures of central tendency and variability, sound engineering and TS, to automatically select the best descriptors in a given audio context. In general, several libraries have been developed to automatically generate different features for TS such as TS-Features [15, 16] and TS-fresh [6], which have been used in various articles about TS forecasting [1–3, 13, 14, 24, 38].

Historically, the prediction of seasonal data has received significant research efforts, and in recent decades, many theoretical methods have been developed for its modeling. Among the most frequently used statistical methods for sales forecasting are Exponential Smoothing (ES) and its extensions (Holt, Pegels, and Holt-Winters) [10]. This method averages (smoothes) TS data, but unlike a simple moving average, it assigns more weight to recent observations and exponentially decreases the weight of observations over time. Like ES models, the autoregressive integrated moving average (ARIMA) models are recognized as one of the most widely used methods for TS forecasting [2].

In recent years, ML methods have been proposed as an alternative for forecasting TS [8, 13, 25], due to their ability to identify non-linear patterns and the few assumptions that the data must meet for a correct implementation [5]. The most popular models come from Artificial Neural Networks (ANN). Various types of ANN have been developed to estimate demand in the retail sector, such as Long Short Term Memory Neural Network - LSTM, which is considered one of the best methods for TS forecasting [22]. Lui et al. [23] use an Economic Order Quantity (EOQ) model for inventory control in manufacturing systems. The cost of the inventory-production system was reduced using game theory and fuzzy demand forecasting. The paper of

Giri and Chen [11] defines a smart forecasting system that uses image feature attributes of clothes along with its sales data to estimate future demands. This forecast model is based on deep learning techniques, which extract the features of the product images to predict weekly sales in clusters of products. In [39], Yu-Chung et al. developed a demand forecasting approach based on TS and ML techniques that selects the best-fitting model as a baseline, and then used Google Trends and a web crawler to collect market signals to incorporate them into the selected baseline model to adjust forecasting results to account for demand fluctuations.

In addition to the statistical and ML techniques, it is important to analyze the ensemble learning models that combine different ML techniques. For example, in the M4 competition held in 2018, ensemble forecasting models were the big winners, occupying the first 12 places out of 17 in the competition [4]. In the aforementioned competition, Jaganathan & Prakash [18] experimented with simple combinations of forecasts (ES, ARIMA, Temporal HIErarchical Forecasting - THIEF, naive model, TBATS, Theta, among others) with equal weights based on the trimmed mean, the median and mean, to implement ensemble forecasts. Pavlyshenko [31] used a stacking approach to forecast the sales from the "Rossmann Store", from a public dataset that is available on Kaggle. Ulrich et al. [40] define an automated model selection framework for retail demand forecasting. For that, they define labeled training data based on the models' performances in previous demand periods with similar demand features. Then, they automate the selection of the most promising model via classification based on the labeled training data.

There are a large number of proposed models to improve the accuracy of demand estimation in the retail industry. However, most studies have aimed to propose a single forecasting method that is used for all sales TS under their study [8, 10, 28, 31, 35]. Among the articles that have proposed forecasting methods according to the different features of the product TS, is the one presented by Spiliotis et al. [36], in which used a previous product categorization and evaluated eleven statistical and seven ML methods for a set of 3,300 products from a retail company in Greece. The most recent article, to the best of our knowledge, that conducted a comparative forecasting model study for different product clustering, is the one proposed by Kiefer et al. [21]. The study used a categorization in which the products are segmented according to their intermittency and variability.

3 Our proposal

Our system is defined by seven steps described below (see Fig. 1):

1 Exploratory data analysis: Obtaining an understanding of the dataset based on the identification of the variables, the statistical analysis of central and dispersion measures, the detection of anomalous data, etc.
2 Feature extraction from product TS: Developing an autonomous feature extraction process to capture different variables that are associated with the demand of the products. The idea is to generate variables that could explain the cyclical, trend, seasonal, and irregular behavior of the TS.

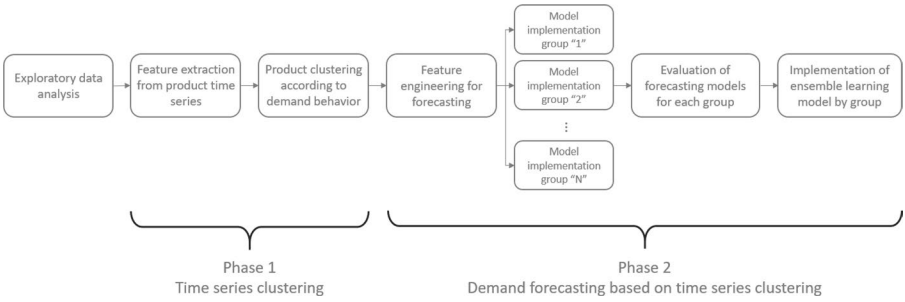


Fig. 1 Our proposal

3 Product clustering according to demand behavior: Developing a product clustering process based on similar features of the demand behavior that are identified in the TS.
4 Feature engineering for forecasting: This step is composed of three processes: First, the feature extraction process defines the input features (X) that relate to the output (Y). Second, a filtering process where strongly correlated features will be eliminated. Finally, the feature selection process, where the most important input variables will be selected according to how useful they are at predicting the target variable. This process will be done with each cluster defined in the previous phase.
5 Model implementation by cluster: This step will select different forecasting methods for each product cluster. In that way, different statistical and ML models will be implemented.
6 Evaluation of forecasting models for each cluster: Several quality metrics will be selected to evaluate the performance of all the implemented models in each cluster of products. Particularly, the next metrics will be considered: R2, MSE (Mean Squared Error), and Mean Absolute Scaled Error (MASE).
7 Implementation of an ensemble learning model by cluster: Different ensemble learning models will be implemented for each product cluster, based on the models with the best performance in the previous step.

4 Experimentation

4.1 Dataset and case study

The dataset involves the unit sales of various products sold by a major retailer in the USA, organized in the form of clustered TS [25, 26]. More specifically, the dataset has 3,049 products, classified into 3 product categories (Hobbies, Foods, and Household). The products are sold across ten stores, located in three states (CA, TX, and WI). The historical data range from 2011-01-29 to 2016-05-22. The objective of the case study is to forecast the last 28 days of the dataset (4 weeks). That is, estimating the demand for all the products.

4.2 Application of our approach

4.2.1 Exploratory data analysis

4.2.1.1 General analysis of the dataset The aggregated sales of the products show that the average demand is 532 units per day per store. It is also important to mention that there is a cyclical behavior in annual sales. Sales for the year start with a progressive decrease in unit sales from January to April, and then an increase in demand begins in September. Finally, the sales start to decline again from September to next year's April. Figure 2 is a smoothed graph with a 90-day moving average so that the described phenomenon can be more easily appreciated.

4.2.1.2 Descriptive analysis by category The category with the highest average demand per day and the highest variance is the "Foods" category. Regarding the "Household" and "Hobbies" categories, they have a similar average daily demand, which is around 40 units sold per day. The above information is relevant to the proposed forecasting system, since products of the "Hobbies" and "Household" categories will have the highest product proportion with intermittent demand, and consequently, will be the most difficult products to forecast.

4.2.1.3 Sales behavior in temporary variables It is important to understand how the product categories behave in different time windows to include date-time features as input information for the forecasting models. Here are the most relevant findings:

- Day's analysis: the three categories have very similar behavior in the average demand sales of the days of the week. As expected, sales have their highest peak on weekends, while for weekdays demand remains at the lowest levels (Except for Friday, where demand begins its growth due to being the weekend eve).
- Month's analysis: the "Household" category has two sales peaks throughout the year, one around April and the other in August. Regarding the "Hobbies" category, the highest peak of sales is in April and it notably drops in May. Additionally, a decrease in sales can be observed for the months that coincide with the coldest

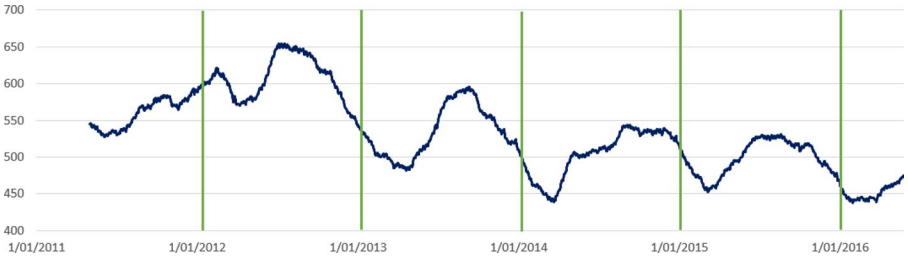


Fig. 2 Moving average (90-days) of the aggregated sales

seasons of the year. Finally, regarding the "Foods" category, its sales begin to increase from April.

- Year's Analysis: Regarding the "Hobbies" category the unit sales tend to be higher in odd years than even years. The "Household" category seems to be recovering from a sales drop in 2014 and is having the best daily average sales in 2015 and 2016. Lastly, the "Foods" category has a negative trend that started in 2013.

4.2.2 Feature extraction for product clustering

Different features were extracted from each product's TS. The main objective is to generate variables that could explain the cyclical, trend, seasonal and irregular behavior of the TS to later cluster the products based on the extracted features. In this way, the demand of every product was analyzed to extract features like peaks, trends, and entropy, among others, to identify and describe the components of each TS. At the end of the feature extraction process for each product's TS, 44 variables described different aspects of the TS. Due to the high number of variables, an elimination process of correlated variables was carried out so that there were no variables that would explain similar phenomena of the TS. For this, the Pearson correlation was used and it was defined that variables with greater values than a defined correlation threshold were filtered (For this work, 0.8). The resulting variables with which the product clusters will later be generated are shown in Table 1, specifying whether its origin is statistical (ST) or TS.

4.2.3 TS clustering

Two different clustering methods were implemented: K-means and hierarchical clustering. For each clustering method, two alternatives were evaluated: A scenario using all the variables selected in the previous step and another using PCA. Hierarchical clustering and K-means methods, which are distance-based methods, were selected over other density-based methods as they ensure that all products are assigned to a cluster. Regarding the evaluation metric, it was used the silhouette metric because is also based on distances between the elements and is a self-contained metric (that is, depending on the value it takes, the quality of the grouping result can be known (it varies from -1 for overlapping clusters to 1 for defined and separate clusters). Table 2 shows the tests carried out for different cluster numbers, both for K-means and hierarchical clustering, using all the variables of the dataset or PCA.

The results obtained in Table 2 using the silhouette metric show that the suggested number of clusters is 4 for each of the evaluated methods. Since the best result was obtained with K-means and PCA, it was decided to use the clustering obtained by this case during the rest of this work. After the cluster definition process, a study of the average variables of each cluster of the selected clustering was carried out (see Table 3). The objective was to understand the general properties of the products that composed each of the clusters. Table 3 shows some of the main variables of each centroid of each cluster.

Table 1 Description of the extracted features selected

Feature	Description	Type
Autocorrelation	Calculates the value of an aggregation function f_{agg} (e.g. the variance or the mean) over the autocorrelation R(l) for different lags.	TS
count_above_mean	Returns the number of values in x that are higher than the mean of x.	ST
Kurtosis	Returns the kurtosis of x (calculated with the adjusted Fisher-Pearson standardized moment coefficient G2). Kurtosis measures the peakedness of a TS.	ST
linear_trend	Calculate a linear least-squares regression for the values of the TS versus the sequence from 0 to the length of the TS minus one.	ST
longest_strike_above_mean	Returns the length of the longest consecutive subsequence in x that is bigger than the mean of x.	ST
mean_change	Returns the mean over the differences between subsequent TS values.	ST
Median	Returns the median of x.	ST
number_peaks	Calculates the number of peaks of at least support n in the TS x. A peak of support n is defined as a subsequence of x where a value occurs, which is bigger than its n neighbors to the left and the right.	TS
sample_entropy	Calculate and return sample entropy of x. Used for assessing the complexity of physiological time-series signals, and diagnosing diseased states.	ST
Skewness	Indicates the symmetry of the probability density function (PDF) of the amplitude of a TS.	ST

Table 2 Silhouette metric results for different instances of methods vs. input variables

Number of clusters	K-means		Hierarchical clustering	
	All variables	PCA	All variables	PCA
2	0,1958	0,2601	0,1804	0,2395
3	0,2106	0,2751	0,1874	0,2595
4	**0,2334**	**0,3123**	**0,2078**	**0,2793**
5	0,2018	0,2463	0,1396	0,2476
6	0,1961	0,2961	0,1461	0,2791

In Cluster 0, the products are characterized by high intermittency. On the contrary, Cluster 1 is characterized by a continuous demand, and also, has the lowest null demand cases in its TS. Additionally, this cluster is the only one with a negative trend. Cluster 2 stands out for having the lowest auto-correlation value against the demand of the previous weekday. Cluster 3 has similar features to Cluster 0. However, it has a lower presence of null demand and a much higher number of demand peaks.

Table 3 Average of the main variables of the centroid of the clusters of the selected clustering

Cluster	Kurtosis	Mean	Auto-correlation lag 7	Number peak n_6	Linear trend "slope"	Value count 0
0	13,4	1,5	0,33	131	0,000	1.050
1	2,8	17,0	0,48	198	-0,004	225
2	5,8	4,5	0,19	183	0,000	473
3	13,3	2,4	0,26	170	0,000	830

Table 4 Type of features for the forecasting models

Feature class	Description
Date Time Features	Information from the date/time values of each observation
Lag Features	Values at prior time steps
Window Features	Summary of values over a fixed window of prior time steps.

4.2.4 Feature engineering for forecasting

A feature engineering process was developed to provide strong relationships between the input features and the real demand value. The choice of the most appropriate set of features depends on the nature of the analyzed TS. In consequence, this work implemented three different classes of features. Table 4 provides a brief description of the features used in this work.

After the feature extraction process, each product ended up with 93 variables that described different demand behaviors. Therefore, a feature selection process was needed to choose the best variables for each of the product clusters. This process was done in two steps. First, a Pearson correlation process was implemented to eliminate variables with a higher correlation from a defined threshold (For this work, 0.8). After, a feature selection process based on Random Forest was implemented. In this way, this work left those features with a higher coefficient than a defined threshold (the project used a 0.01 threshold).

4.2.5 Model implementation

This work implemented several forecasting methods based on statistical, ML and deep learning models. The implemented forecast models used in the work are mentioned below. All the parameters for each technique were the result of a parameter tuning process using the grid search hyperparameter optimization strategy [29, 31].

Naïve Setting all forecasts to be the value of the last observation [15]. In this way, the forecast at time t, $\widehat{y}_t$, is equal to the last known observation of the TS, and $y_{t\text{-}1}$ would be the product demand of the previous day.

$$\widehat{y}_t = y_{t-1}$$

Naive – previous weekday value Setting all forecasts to be the value of the last weekday observation. In this way, the forecast at time t, $\widehat{y_t}$, is equal to the last weekday observation of the TS, y, as follows.

$$\widehat{y_t} = y_{t-8}$$

For example, the prediction of the next Wednesday's product demand would be the demand of the previous Wednesday.

Moving average The forecasts of all future values are equal to the average (or "mean") of the historical data [15, 16]. Let the historical data be denoted by $y_1, y_2, ... y_t$, then the forecast could be defined as.

$$\widehat{y_t} = (y_1 + y_2 + \ldots + y_{t-1})/(T-1)$$

Two periods were implemented for the case study. One with the last 7 observations, and the second with the last 28 demand observations.

Triple exponential smoothing Forecasts produced using ES methods are weighted averages of past observations, with the weights decaying exponentially as the observations get older [15]. Triple Exponential Smoothing is an extension of ES that explicitly adds support for seasonality to the univariate TS.

AUTOARIMA it was chosen for this case study because it helps to identify the optimal parameters for an ARIMA model. AUTOARIMA emerges as a relevant option to adapt the different model parameters to each product.

Linear regression it fits a linear model with coefficients $W = w_1, w_2, ..., w_3$ to minimize the residual sum of squares between the observed targets in the dataset, and the targets predicted by the linear approximation.

KNN Regression KNNR is a similarity-based method, generating forecasts according to the Euclidean distance computed between the points used for training and testing. Given x_i points as inputs, the method picks the closest k points of the training sample to them and then sets the prediction equal to the average of their corresponding target values.

Random Forest it is a combination of Regression Trees, each one depending on the values of a random vector sampled independently and with the same distribution.

Gradient boosting trees (GBT) GBT builds one tree at a time, each new tree correcting the errors made by the previously trained one.

Long short-term memory neural networks (LSTM) LSTM as an extension of RNN has a strong capability in forecasting TS data. LSTM can store long-range time dependency information and can suitably map between input and output data [12].

The model that was implemented in the case study was a Stacked LSTM model, composed of two hidden LSTM layers. The first layer had 32 units while the second layer had 16 units. The output layer was a dense layer with one unit.

4.2.6 Evaluation of forecasting models for each cluster

The quality metrics MSE, R^2 and MASE were selected for the performance evaluation of the forecasting models. MSE is one of the most popular functions in demand forecasting. On the other hand, R^2 is a self-contained metric since its maximum value is 1. Finally, the MASE metric was selected since it compares the result of the forecast model with the naive model. In this way, if MASE is greater than 1, then it means that the model has worse performance than the naive model. Table 5 shows the quality metrics results for the four product clusters with all the forecasting models.

According to Table 5, the top-three performing methods in terms of R^2, MSE, and MASE are Linear Regression, LSTM, and Random Forest. Interesting findings are made when examining the results in terms of the product clusters. Cluster 0 showed that Linear Regression and LSTM models had a similar performance in the quality metrics results. However, this cluster achieved the lowest result because of the null-demand recurrency. Regarding cluster 1, the results showed a superiority of the linear regression model over LSTM and Random Forest methods. This cluster had the highest score with the three evaluated metrics, which means that a high mean demand value combined with a high autocorrelation, helps the models to achieve better results. For clusters 2 and 3, the best performance was obtained by the LSTM model. These clusters were the ones with the lowest Autocorrelation value, which probably means that LSTM models have an advantage in products with this feature.

4.2.7 Implementation of an ensemble learning model by cluster

Initially, 3 basic ensemble models were implemented. The first is based on the mean of the 2 best performing models in each cluster; the second is based on the mean of the 3 best performing models in each cluster; and the last is based on the median of the 3 best performing models in each cluster. Subsequently, two blending models were implemented. The first one uses the two best-performing models of each cluster as the base model, and the second one uses the three best-performing models of each cluster as the base model. The base models were trained with the demand data from 01/29/2011 to 03/27/2016 and then, the predictions were made for the validation set between 03/28/2016 to 04/24/2016. From the obtained forecast of the base models in the validation set, a Linear Regression model (Meta-Model) was trained to predict the demand of the test set (Dates between 04/25/2016 to 05/22/2016). Table 6 shows the test results of each of the ensemble models, as well as the obtained result of the best model of each cluster in the sixth step of the methodology.

Table 7 highlights the best-performing models for each of the product clusters. For cluster 0, a very similar performance was observed between the blending and the mean model using the 2 best models in step 6. However, the blending model was selected as the best one in cluster 0 because it obtained better results in 2 of the 3

Table 5 Results in each product cluster for all the implemented models

Method	Cluster 0			Cluster 1			Cluster 2			Cluster 3		
	R^2	MSE	MASE	R^2	MSE	MASE	R^2	MSE	MASE	R^2	MSE	MASE
Naive	-0.98	6.49	1.53	-0.63	63.13	0.82	-0.76	25.58	0.98	-0.77	8.56	1.27
Naive7	-0.93	6.55	1.51	-0.81	58.01	0.86	-0.94	25.81	1.06	-0.96	8.76	1.37
MA7D	-0.18	3.94	1.29	-0.17	46.27	0.74	-0.17	16.66	0.87	-0.19	5.67	1.11
MA28D	-0.07	3.54	1.21	-0.10	45.31	0.72	-0.08	15.70	0.83	-0.07	5.09	1.03
TES	-0.10	3.54	1.19	-0.33	38.58	0.73	-0.09	15.82	0.82	-0.13	5.04	1.00
ARIMA	-0.22	4.08	1.23	-0.46	52.12	0.82	-0.20	18.18	0.86	-0.22	5.71	1.06
LR	**0.19**	**2.39**	**1.08**	**0.46**	**24.37**	**0.49**	0.20	10.64	0.71	0.09	4.09	0.95
KNN	-0.20	3.81	1.26	-0.09	39.32	0.69	-0.22	15.47	0.86	-0.18	5.55	1.04
RF	0.02	3.05	1.20	0.14	34.10	0.62	0.08	12.00	0.78	0.03	4.45	0.99
LGBM	-0.04	3.36	1.25	0.03	37.80	0.65	0.03	12.10	0.79	-0.05	4.88	1.03
LSTM	0.18	2.64	1.06	0.42	29.57	0.53	**0.22**	**9.97**	**0.69**	**0.19**	**3.65**	**0.89**

Table 6 Results of each of the ensemble models and the best model for each cluster

Method	Cluster 0			Cluster 1			Cluster 2			Cluster 3		
	R^2	MSE	MASE	R^2	MSE	MASE	R^2	MSE	MASE	R^2	MSE	MASE
Best_M	0.197	2.391	1.080	0.458	24.36	0.496	0.221	9.969	0.686	**0.191**	**3.650**	**0.888**
CMean2	0.224	2.394	1.052	**0.481**	**24.67**	**0.494**	0.238	9.78	0.684	0.180	3.704	0.910
CMean3	0.186	2.533	1.094	0.397	26.88	0.529	0.204	10.18	0.711	0.151	3.867	0.928
CMedi3	0.175	2.590	1.100	0.402	26.72	0.521	0.196	10.41	0.714	0.131	3.984	0.942
Blend2	**0.186**	**2.349**	**1.038**	0.466	26.10	0.508	**0.245**	**9.397**	**0.678**	0.163	3.688	0.862
Blend3	-1.76	3.918	1.366	0.453	25.61	0.499	0.196	9.914	0.689	0.127	4.068	0.899

quality metrics (MSE and MASE). For cluster 1, the combined model with the mean of the two best models was the one that obtained the best performance of all the proposed alternatives. However, the blending model with 2 base models also got better performance than the best method in step 6. Regarding the products of cluster 2, it was observed that the best model was the blending one with 2 base models since it got better results in each of the three quality metrics than the other proposals. Finally, in cluster 3 no ensemble model obtained better performance than the LSTM method. One of the likely reasons is the wide difference between the LSTM performance with the other implemented models. Table 6 shows that LSTM has an R^2 greater than twice the second model with the best performance in cluster 3 (Linear Regression). In general, the ensemble models have better behavior because they have the adaptive capacity to choose the best results as the final result.

4.3 Results analysis

According to Table 6, the demand behavior influences the model performance. The product clusters of the proposed forecasting system had better results with different types of forecasting models. Particularly, the products from Cluster 0 and Cluster 1, which were characterized by a high autocorrelation value, had a better performance with the linear regression model, while the products of Cluster 2 and Cluster 3, which were characterized by a low autocorrelation value, achieved better results with the LSTM model. However, the best forecasting results were derived from the ensemble models that combined the best-performing models of each cluster. In this case, the blending model and the combined mean model that used the LSTM and Linear Regression methods obtained better results than the individual models in Cluster 0, Cluster 1, and Cluster 2. Regarding Cluster 3, none of the ensemble models exceeded the performance of the best individual model. This likely happened because the LSTM model significantly outperforms the other individual models in the quality metrics result. The difference in R^2 between LSTM and the second-best model in Cluster 3 is 0.1. This does not happen in any of the other clusters where the ensemble models had better performance.

In general, the combination of the clustering product process and the use of ensemble models based on the top-performing models of each cluster generates better results than the individual models. Table 8 shows the three quality metric comparisons between the proposed forecasting system and the individual models used in the sixth step. The proposed forecasting system considers the best model for each cluster to calculate the overall result. The quality metrics used to obtain the best model for each cluster are R^2, MSE and MASE (highlighted in Table 6). For each model, the overall average for the quality metrics of all products is shown in Table 7.

Note that the proposed forecasting system obtained better quality metrics results in each of the evaluated measures. Comparing the proposed approach with the top individual forecasting model (LSTM) shows that the R^2 measure is 9.32% higher, the MSE value is 10.21% lower, and the MASE is 2.18% lower. Additionally, a second comparison is proposed in which the general average of the quality metrics is calculated according to each product and its best individual model according to the cluster

Table 7 The general results of the proposed forecasting system and the individual models

Model	R^2	MSE	MASE
Our approach	**0.2592**	**9.3262**	**0.7573**
LSTM	0.2371	10.3873	0.7742
Linear Regression	0.2250	9.9916	0.7873
Random Forest	0.0740	12.2687	0.8804
Gradient Boosting Trees	0.0054	12.9535	0.9058
Moving Average (28 last days)	-0.0817	15.9731	0.9232
Triple Exponential Smoothing	-0.1307	15.0921	0.9162
Moving Average (7 last days)	-0.1786	16.7641	0.9731
KNN Regression	-0.1935	15.1222	0.9469
ARIMA	-0.2446	18.4267	0.9630
Naive	-0.7920	24.7241	1.1172
Naive Previous Weekday	-0.9232	24.1600	1.1704

Table 8 Comparison between our model with a composed model from the best model of each cluster

Model	R^2	MSE	MASE
Our approach	**0.2592**	**9.3262**	**0.7573**
Best individual model of each cluster	0.2457	9.5975	0.7706
Percentage of improvement	**5.49**	**-2.82%**	**-1.73%**

Table 9 Performance of our approach in forecasting task

Model	MSE	MASE
Our approach	11.1451	0.7611

to which it belongs. Finally, the comparison of the "best individual model of each cluster" and our proposal shows its contribution (see Table 8).

Our proposal obtained better quality metrics in each of the measures in this second comparison. Table 8 shows that the R^2 measure is 5.49% higher, the MSE value is 2.82% lower, and the MASE is 1,73% lower.

Finally, a last test was done in which 80% of the TS was taken for training and 20% of the TS was left to evaluate its quality. In that case, cross-validation was used so that interleaved pieces were taken to predict (only the case to predict the beginning of TS was not considered). Table 9 shows the average result obtained using the MSE and MASE metrics. We see that in general. The behavior of our approach follows the results achieved in Tables 7 and 8, that is, it does not degrade its performance when carrying out forecasting processes.

5 Comparison with other works

5.1 Quantitative analysis

In order to carry out a comparison of the quality of the results obtained with our approach, we have identified the previous works that propose a dataset that can be accessed, report the results obtained, and automate the analysis of TS that describes clusters of products to build ensemble ML models specific to each one of them.

Table 10 Comparison between our model with a composed model from the best model of each cluster

Work	Technique	R^2	MSE	MASE
[1]	ARIMAX	NA	NA	0.337
	Our approach	**0.902**	NA	0.321
[11]	CNN	NA	0,038	NA
	Our approach	**0.851**	0.041	0.315
[21]	LSTM	NA	NA	0.980
	Our approach	**0.873**	NA	0.401
[40]	GAMLSS	NA	0.024	NA
	Our approach	**0.892**	0.032	0.342

Table 11 *P*-values: volatility impact on the forecasting accuracy (MASE)

Dataset	Result
[1]	0.124
[11]	0.170
[21]	0.135
[40]	0.126

Table 10 describes the results obtained with our approach with the selected datasets, and those obtained with the original work.

The first dataset is from Abolghasemi et al. [1]. This dataset has hundreds of demand TS of a major fast-moving consumer goods food manufacturing company in Australia. In that work, they only calculated MASE, and the best metric was ARIMA with covariate (ARIMAX). Our approach has very good behavior, with a similar MASE to this work and an R^2 very good. The second dataset used is from a European fashion retail company for a span of two years (2015, 2016), which consists of sales information of 290 items [11]. This second work, like ours, makes a clustering of products, and for each of them is carried out the demand forecast models (the result shown in Table 10 is the average quality of the models). In this case, again our approach has very good behavior, with an R2 close to 1. The third dataset is one used on the M5 Competition, which contains intermittent and lumpy TS. The dataset is from Walmart and has around 100,000 daily TS [21]. The last dataset includes 546 TS that covers the categories of fruits, vegetables, and bake-off products from five regional fulfillment centers [40]. The best model was the Generalized Additive Model for Location, Scale, and Shape (GAMLSS). In all these datasets, our approach has very good results with R^2 close to 1, and with MASE or MSE values close to the compared work. This comparison shows the ability of our approach to be used in different datasets, always obtaining results that are among the best, if compared with other ML methods (see Tables 7 and 8, and 10).

Additionally, we used pairwise t-tests to determine whether the forecasting accuracy (MASE) of our model significantly differs when series show different levels of volatility (see Table 11). In general, we can see that our approach is a very robust approach that performs well across different datasets with different levels of volatility. There is no statistically significant difference in the accuracy.

5.2 Qualitative analysis

The comparison of the proposed forecasting system with other studies was done with the following criteria (see Table 12): (i) Implemented forecasting models: The forecasting techniques that were used in the study; (ii) Product clustering method: The method with which the products were categorized; (iii) Ensemble Models: If ensemble learning models were evaluated in the study.

Spiliotis et al. [36] evaluated the performance of Statistical and ML methods for 3300 real-TS of various consumer goods sold by a major retailer in Greece. The results showed that ML methods can provide significantly less biased and more accurate forecasts than well-established, statistical methods, like Croston's method and its variants. Particularly, the results showed that for intermittent products the Random Forest models, GBT, SVR and the Syntetos-Boylan method were the best models in terms of RMSSE; for the lumpy category, the best methods were the four previous models and KNNR; regarding the smooth products, the best methods were GBT and Random Forest; and finally, for the erratic class, the top-performing models were

Table 12 Comparison with other works

Work	Implemented forecasting models.	Product clustering method.	Ensemble models
Our approach	Naïve, Seasonal Naïve, Moving Average. Triple Exponential Smoothing, Auto-ARIMA, Linear Regression, KNNR. Random Forest, Gradient Boosting Trees, LSTM.	K-Means with uncorrelated demand features.	Yes
[11]	Croston, Triple Exponential Smoothing. Auto-ARIMA, Random Forest, XGBoost, Auto-SVR, MLP, LSTM.	Williams' categorization.	No
[36]	Naïve, Seasonal Naïve. Simple Exponential Smoothing. Moving Averages, Croston's method, Syntetos–Boylan Approximation. Shale–Boylan–Johnston Approximation, Teunter–Syntetos–Babai method, ADIDA, iMAPA, Multi-Layer Perceptron (MLP), Bayesian Neural Network, Random Forest, Gradient Boosting Trees. KNNR, Support Vector Regression, Gaussian Processes (GP).	Syntetos et al. categorization.	No

SVR, Random Forests, GBT and KNNR. Kiefer et al. [11] used Williams's demand categorization to compare statistics and ML methods by applying a metric, called Stock-keeping-oriented Prediction Error Costs (SPEC). The results evidenced that the Croston algorithm is well suited to demand forecasting of intermittent and lumpy TS because it was the top-performing model in each of the product categories.

Based on Table 12, the proposed forecasting system differs from the other articles because it proposes a dynamic clustering process based on different uncorrelated demand features that describe the cyclical, trend and seasonal components of the TS. In this way, the proposal selects the number of clusters according to the silhouette score of a K-Means algorithm, while the other studies use fixed values on only two types of variables that describe the intermittence and lumpiness of the TS. In consequence, the proposed system can cluster similar products that could be categorized in the same cluster by having a similar value on the two variables that are examined in the fixed clustering proposal. Additionally, the forecasting system that is proposed evaluates the implementation of ensemble learning models from the top-performing forecasting methods of each cluster. This is not evaluated in the other studies and demonstrates that the demand estimation is enhanced when the base models that are used for the ensemble model have similar performances in the quality metrics. Particularly, the main contribution of this work, with respect to previous ones, is that it defines the most appropriate forecasting models for product categories in real time. For that, our approach carries out an online product clustering with a feature engineering process for TS included; and has several forecasting models by product category (cluster) that are integrated using ensemble learning models in an autonomous demand forecasting system.

6 Conclusions

Limited research has been done about demand forecasting based on the comparison of different methods in clusters of products with distinct TS features. Additionally, to the best of our knowledge, there are no studies evaluating ensemble models within the framework of product clustering according to the features of the demand. In this way, this study seeks to contribute by (1) proposing a novel product clustering process based on a dynamic feature selection process according to the demand behavior of the selected products, (2) identifying the best forecasting models for each cluster of products from 10 different method alternatives, (3) evaluating the estimation benefits of ensemble models from the top-performing methods of each of the composed clusters. The major benefit of this two-step strategy (cluster first, then forecast) compared to a one-step strategy (using features to forecast in a classification/regression manner) is that it allows the definition of forecasting models by product category, which allows following the very diverse behavior that exists between the products and adapting to new behaviors.

The results showed that the proposed forecasting system can select the number of product clusters according to the demand features that are associated with the TS. In this way, an aggregation of four different clusters was recommended. Additionally, it is observed that after the feature engineering process, each of the clusters has its

feature importance ranking and a different set of top-performing models, being linear regression and LSTM the best forecasting models in every cluster of each scenario. However, it was identified that according to the cluster variables, the linear regression model performed better with products with a high demand autocorrelation, while LSTM had better results with low autocorrelated demand.

Regarding the ensemble models that were implemented in the proposed forecasting system, it is observed that they are an important alternative to take into account when the base forecasting models have acceptable and similar performances in the quality metrics. Thus, the mean and blending models that used linear regression and LSTM as base models were the top-performing models in clusters 0, 1 and 2 since the difference in their R^2 was less or equal to 0.04. On the other hand, the mean and blending models that used base models with high-performance differences obtained lower results compared to the best individual model.

Different aspects are worth studying for future work. First, it would be interesting to integrate more variables from external information related to this case study, such as macroeconomic and weather variables, to analyze if they could be useful to achieve better results. Another aspect to focus on is the evaluation of other forecasting models, like Prophet or Croston, to identify those with similar or better performance than linear Regression and LSTM. This is for extending the knowledge about which models suit better to each cluster of products and using them as base models for the ensemble model process. Regarding the clustering product process, it would be interesting to compare our clustering process against Williams' and Syntetos' clustering method with different sets of products to evaluate its influence on the proposed forecasting system performance. At the same time, as blending models allow the use of different options of meta-models (a model that learns how to best combine the predictions of the base models), it would be interesting to examine other different options from the linear regression to evaluate if the blending models could reach better results. Finally, we see that, in general, the results obtained with our approach are quite good. In the original dataset with which the behavior of our model was detailed (see Sect. 4), we obtained the worst results, but it is the best among the ML techniques tested with that dataset. Future work will use a classification of products different from how they are labeled in this dataset, to determine if the classification into 3 product categories (hobbies, food, and home) influences the quality of the results.

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Data availability Data are available in [25, 26].

Declarations

Competing interests The authors declare no competing interests.

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